

System Boot Process

The **system boot process** is the sequence of steps a computer follows to start the operating system after power is turned on.

The term **booting** means:

Loading the operating system kernel into memory and starting it.

Basic Steps in System Boot

The boot process generally follows these stages:

Power On

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Bootstrap Program / Boot Loader

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Load Kernel into Memory

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Kernel Initializes Hardware

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Mount Root File System

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Start System Services

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Login Screen / Desktop

1. Bootstrap Program (Boot Loader)

When the computer is powered on, a small program called the:

- **Bootstrap program**
- or **Boot loader**

starts executing.

Its job is to:

- Locate the operating system kernel
- Load it into RAM
- Start kernel execution

Examples of boot loaders:

- GRUB (Linux)
 - LK
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BIOS-Based Boot Process

Older systems use **BIOS** (Basic Input/Output System).

BIOS Boot Steps

1. BIOS firmware starts
2. BIOS loads boot code from the disk's **boot block**
3. Secondary boot loader loads the kernel
4. Kernel starts execution

Because the first-stage loader must fit in a small disk sector, it is very simple.

UEFI Boot Process

Modern systems use:

UEFI (Unified Extensible Firmware Interface)

Advantages over BIOS:

- Faster booting
- Better support for:
 - 64-bit systems
 - Large hard disks
- More advanced boot management

Unlike BIOS, UEFI can directly function as a complete boot manager.

Kernel Loading

The boot loader loads the kernel image into memory.

Example Linux kernel parameter:

```
BOOT_IMAGE=/boot/vmlinuz  
root=/dev/sda1
```

Here:

- BOOT_IMAGE → kernel file
- root → root file system location

Linux kernel images are usually **compressed** to save space and reduce boot time.

Hardware Initialization

After loading, the kernel:

- Detects CPU
 - Detects memory
 - Initializes device controllers
 - Loads drivers
 - Configures hardware
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initramfs (Initial RAM File System)

Linux temporarily creates a small RAM-based file system called:

initramfs

Purpose:

- Store essential drivers/modules
- Help mount the real root file system

Flow:

Temporary initramfs

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Load Drivers

↓

Mount Real Root File System

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Switch Root

After switching:

- Traditional Linux removes initramfs
- Android keeps it as the main root file system

Root File System

The **root file system** is the main directory structure of the OS.

It contains:

- System files
- Libraries
- Applications
- Configuration files

Mounting the root file system means making it accessible to the kernel.

Starting systemd / init

After the kernel setup finishes, Linux starts the first user-space process:

- Traditionally: init
- Modern Linux: systemd

systemd

This process:

- Starts background services
 - Launches networking
 - Starts databases/web servers
 - Opens login screen
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Android Boot Process

Android differs slightly from desktop Linux.

Android:

- Uses Linux kernel
- Does not use GRUB
- Commonly uses LK
- Keeps initramfs as the permanent root file system

After booting:

1. Kernel loads
2. Root file system mounts
3. Android init process starts
4. Services launch
5. Home screen appears

Recovery Mode / Single-User Mode

Most operating systems provide special boot modes for troubleshooting:

- Recovery mode
- Single-user mode
- Safe mode

These are used for:

- Repairing file systems
- Diagnosing hardware
- Reinstalling OS
- Fixing boot problems

Supported by systems such as:

- Linux
- Windows
- macOS
- Android
- iOS

Summary

Component	Function
Boot Loader	Finds and loads kernel

Component	Function
Kernel	Initializes hardware and system
initramfs	Temporary RAM-based root file system
Root File System	Main OS directory structure
systemd/init	Starts system services

In Short

Booting is the process of loading the operating system kernel into memory, initializing hardware, mounting the root file system, and starting system services until the computer becomes ready for use.